End – Semester Examination (Spring Semester) – 2023-24
[B.Tech, CSE – 3rd Year, 6th Semester]

Paper Name: Data Science & Big Data **Full Marks: 80**
Paper Code: CSEUGPE02 **Time: 3 Hrs**

GROUP A
[Answer any *five* questions] [5 x 2 = 10]

1. i) How many association rules can be produced from a k-itemset?
ii) What do you understand by downward closure property of Apriori Principle for the itemsets?
iii) What data transformation is required for ratio-scaled attributes to apply Euclidean distance?
iv) Why data normalization is needed?
v) How min-max normalization differs from the standardization (z-score).
vi) Distinguish between unsupervised and supervised learning.

GROUP – B
[Answer any *four* questions] [4 x 5 = 20]

2. Explain the Apriori algorithm for association rule mining.
3. Describe the different steps in Data Mining.
4. Categorize different types of attributes with examples.
5. Explain the following terms in the context of association rule mining: support of an itemset, confidence of an association rule, and frequent itemset.
6. Consider the sample dataset with three features (A, B, C) with target classes 'X' and 'Y': **{(9.6, 88, 8.3, X), (6.2, 47, 5.3, Y), (8.1, 83, 7.3, X), (8.5, 83, 8.1, X), (6.7, 52, 4.2, Y), (8.3, 75, 7.4, X), (5.9, 40, 5.1, Y), (8.9, 92, 9.2, X)}**. Find the class label of the test instance **(6.3, 42, 5.2)** using the K-Nearest Neighbor (KNN) classifier with **k = 4**.
7. Consider the following confusion matrix and compute the values of accuracy, precision, recall, false positive rate and F-measure from it.

	Actually Horse	**Not Actually Horse**
Predicted Horse	150	15
Predicted Not Horse	5	70

GROUP – C
[Answer any *five* questions] [5 x 10 = 50]

8. a) Differentiate between agglomerative and divisive hierarchical clustering algorithms.
b) Given the following distance matrix, create the dendrogram using agglomerative complete linkage clustering.

[3 + 7]

	1	2	3	4	5
1	0.0				
2	2.2	0.0			
3	6.4	5.2	0.0		
4	10.2	9.0	4.3	0.0	
5	9.1	8.0	5.2	3.1	0.0

9. Consider the following dataset {(4,3), (2,1), (3,5), (5,6), (7,8), (6,7)}.
   a) Compute the covariance matrix for the given dataset.
   b) Find the Eigen values and unit length Eigen vectors from the obtained covariance matrix. [4 + 6]

10. a) Reduce the dimension of the dataset provided in Q9 to single dimension through Principal Component Analysis.
    b) Describe the k-means clustering algorithm. [6 + 4]

11. a) Given the following transaction database, find all the frequent itemsets for minimum support value 3 using Apriori algorithm. t1: a, c, m t2: c, t, m t3: a, e t4: a, c, t, e, m t5: c, s t6: a, c, e t7: c, m, t
    b) Define an alternative performance metrics to overcome the limitation of support and confidence metrics in association rule mining. [7 + 3]

12. a) Consider the following transaction database. If minimum support is 3, construct the FP-growth tree from the database. t1: e, s t2: a, d, e t3: d, m, t t4: a, d, m t5: d, t, m t6: a, e t7: a, d, t, e, m
    b) Find the patterns from the obtained FP-tree growth algorithm from the above question. [5 + 5]

13. a) Compute the dissimilarity between the objects based on individual attributes for the dataset given below.
    b) What will be the effective dissimilarity matrix between the objects by considering all the mixed type attributes? [6 + 4]

Object Identifier	*test-1 (nominal)*	*test-2 (ordinal)*	*test-3 (numeric)*
1	M	Excellent	47
2	N	Fair	24
3	P	Good	66
4	Q	Excellent	28

14. Define the entropy measure. Compute the entropy and information gain for all the attributes for the given dataset to identify for the root node in the construction of ID3 decision tree algorithm. [2 + 8]

Sl No	Age	Income	Graduate	Credit Rating	Buys Laptop (Class Label)
1	<35	High	No	Fair	No
2	<35	High	No	Excellent	No
3	35-50	High	No	Fair	Yes
4	>50	Medium	No	Fair	Yes
5	>50	Low	Yes	Fair	Yes
6	>50	Low	Yes	Excellent	No
7	35-50	Low	Yes	Excellent	Yes
8	<35	Medium	No	Fair	No
9	<35	Low	Yes	Fair	Yes
10	>50	Medium	Yes	Fair	Yes
11	<30	Medium	Yes	Excellent	Yes
12	35-50	Medium	No	Excellent	Yes
13	35-50	High	Yes	Fair	Yes
14	>50	Medium	No	Excellent	No

ALIAH UNIVERSITY

End Semester Examination (Spring Semester)2024

(B. Tech 3rd Year 6th Semester)

Subject Name: Software Engineering
Full Marks: 80
Time: 3 Hours
Subject Code: CSEUGPC17

Group A
Answer all *5* questions

1. Answer the following with at most 2 sentences. **5 X 2= 10**
   a) What do you mean by PERT?
   b) What do you mean by cohesion?
   c) What do you mean by coupling?
   d) What do you mean by slack time?
   e) What is the main advantage of Iterative Waterfall Model over Classical Waterfall Model?

Group- B
Answer *any* 6 questions

6 X 5 = 30

2. Explain the concept of Critical Path when scheduling a software project with an example.
3. Give five differences between white box testing and black box testing.
4. Describe Iterative Waterfall Model with a diagram.
5. Suppose the estimated development time and cost using Putnam's expression has come out to be 9 month and ₹10000 respectively. What will be the new cost if we have to develop it within 3 months?
6. Explain why spiral model is also called a Meta model.
7. Explain the Classical Waterfall Model with a diagram.
8. Explain in short prototyping model and evolutionary model. **2.5+2.5**

Group- C

Answer *any 4* questions

4 X 10 =40

9.

a) Explain Alpha, Beta and Acceptance testing.

b) Using black box testing approach find the probable test cases for a software that computes the square root of an input integer which can assume values in the range of 0 to 8000. 5+5

10. Explain the Basic COCOMO model in details. Explain how is Intermediate COCOMO is better than Basic COCOMO? 6+4

11. Write a short pseudo code to find whether a number is prime or not. Draw a CFG and find its cyclomatic complexity. 2+8=10

12.

a) Assume that the size of an organic type software product has been estimated to be 100,000 lines of source code. Assume that the average salary of software developers is Rs. 100,000 per month. Determine the effort required to develop the software product, the nominal development time, and the cost to develop the product.

b) Write five characteristics of a good SRS document. 6+4

13. Explain the different type of classification of cohesiveness and coupling. 5+5

Aliah University

End Semester Examination - 2024

(For 6th Semester B. Tech CSE Programme)

Paper Name: Computer Networks | **Full Marks:** 80

Paper Code: CSEUGPC19 | **Time:** 3 Hrs

Group - A

(10X1=10)

1. Answer any ten (10) questions

   a. What is TTL?
   b. What is the default subnet mask of the ip address 16.192.168.5 ?
   c. What is Ping?
   d. What is the port address of FTP?
   e. What is attenuation?
   f. Which layer of the OSI reference models is responsible for concession control?
   g. What do you mean by 10BASE5?
   h. What is socket?
   i. Give the example of routed protocol
   j. Give the example of **interdomain** routing protocol.
   k. What is subnet mask?
   l. Differentiate between IPv4 and IPV6

Group - B

(Answer any 4 questions) (4X5=20)

2. Briefly describe various types of persistence methods.
3. Differentiate between OSI and TCP/IP reference models.
4. How many types of HDLC frames are there? Describe various types of HDLC frames in detail.
5. Briefly describe Sliding Window protocol. go back N
6. Describe ARP protocol.
7. Describe various types of port address along with their ranges.

Group - C

(Answer any 5 questions) **(5X10=50)**

8. Briefly describe about Pure ALOHA and Slotted ALOHA in detail. **5+5=10**

9. You are given **10100110 bit** stream. Draw the graph using **5X2=10**

a. Manchester coding, b. Differential Manchester

c. NRZ d. NRZ-I e . NRZ-L .

10. You are given **172.16.0.0** ip address slot. You have to create **24** sub networks. **5X2=10**

a. What will be the new subnet mask?
b. What will be the network address of 8^{th} network?
c. What will be the broadcast address of 10^{th} network
d. What will be the ip address of 11th pc of 12^{th} network?
e. What will be the broadcast address of the last network?

11. Briefly describe CSMA/CD with flow diagram. Differentiate between TCP and UDP. **7+3=10**

12. Briefly describe about Leakey Bucket algorithm. Briefly describe 4-way hand-shaking **6+4=10**

13. Describe about TCP header format with a suitable diagram. What is Route poisoning in RIP? **7+3=10**

14. Write short notes (any two) **5+5=10**

(a) RARP
(b) Bellman-Ford Algorithm
(c) TDMA
(d) CRC
(e) ICMP

Aliah University

End-Semester Examination - 2024

(BTech, CSE 3rd Year, 6th Semester)

Paper Name: Compiler Design — **Full Marks: 80**

Paper Code: CSEUGPC18 — **Time: 3 Hrs**

Group – A **(10X1=10)**

1. **Mention whether the statements given below are true or False –**
   a) Left recursive grammars could be used as LL (1) parser.
   b) Type mismatch error is detected during Semantic Analysis phase.
   c) "yytext" is a pointer variable in a lex code.
   d) L-Attributed SDTs can't have Synthesized syntax directed definitions.
   e) DAG representation can form cycle.
   f) Operator grammars never have two consecutive non-terminals.
   g) Peephole optimization technique is needed for optimizing a slice of program at a time.
   h) "aababaab" is a correct string of regular expression=(a(ab)*)
   i) Preprocessor directives are expanded during Lexical analysis phase.
   j) CLR is the most powerful LR parser.

Group – B

Answer any six of the following questions - **(6X5=30)**

2. Discuss the structure of a yacc code. Explain with an example. [5]
3. Discuss Buffer-pair and sentinel concepts used in tokenization. [5]
4. Write a Syntax Directed Translation to count number of '0's in any binary number. Consider any grammar suitable for the same. [5]
5. Differentiate between Bottom-Up and Top-Down parsing techniques. Mention types of each parser. [5]
6. Mention various ways to optimize a loop. Give examples. [5]
7. Write short notes on AST and DAG, and form AST and DAG for the following statement –
   P = (A+B)*C + (A-B)*(A+B) [5]
8. Calculate FIRST () and FOLLOW () of all the non-terminals in the following grammar – [Here S is the start symbol, N={S,A,B,D} and T={a,b,d,e,ε}] [5]

$$S \rightarrow ABaD \mid DbB$$
$$A \rightarrow Ba \mid \varepsilon$$
$$B \rightarrow e \mid \varepsilon$$
$$D \rightarrow d$$

[PTO]

Group - C

Answer any four of the following questions - **(10X4=40)**

9. a) Construct a Non-deterministic Finite Automata (NFA) for the following C-program segment –

```
for (i=0; i<n; n++)
{
    if(a>5 ||b<6)
        p++;
    else
        q++;
}
```

b) Convert the NFA into its corresponding Deterministic Finite Automata. [5+5]

10. a) Write 3-address code for the c-program segment given in question 9(a).

b) Represent the above 3-address code in Quadruples and Triples format. [4+3+3]

11. Consider a grammar (given below) to solve the questions –

$$S \rightarrow L=R$$
$$S \rightarrow R$$
$$L \rightarrow *R$$
$$L \rightarrow a$$
$$R \rightarrow L$$

[Here S is Start symbol. T= {=,a,*} and N={S,L,R}]

a) Find out all LR (0) item sets for the grammar.

b) Form a parsing a table and say whether it is SLR (1) parser or not. [5+5]

12. a) Calculate LR (1) item sets for a grammar given below-

$$S \rightarrow AaAb|BbBa$$
$$A \rightarrow \varepsilon$$
$$B \rightarrow \varepsilon$$

[Here S is Start symbol. N={S,A,B} and T={a,b,ε}]

b) Define Operator grammar. Briefly explain the parsing concept of operator precedence parsing. [5+5]

13. a) What is meant by Peephole optimization? Mention some Peephole optimization techniques with examples.

b) Write semantic rules in the following grammar to evaluate resultant value of any prefix expression – [eg.- +34=7, +2*47=30 etc] [5+5]

$S \rightarrow +SS \mid *SS \mid a$ where T={+, *, a}
